ECEN5692 Digital Communications

Midterm Exam Study Guide — Part 1 (pp. 120–575)

Prepared 3/12/2026 • Four sides of notes + calculator allowed • 75 minutes

Contents

1 Section 1 — Equation Reference Sheet **2**
- 1.1 1.1 Passband / Baseband Signal Representation . . . 2
- 1.2 1.2 I/Q Up and Down Converter . . . 2
- 1.3 1.3 Power Spectral Density (PSD) Relationships . . . 3
- 1.4 1.4 White Noise and Filtering . . . 3
- 1.5 1.5 Complex Baseband Channel (Wireless Multipath) . . . 4
- 1.6 1.6 Doppler Effect and Time-Varying Channels . . . 5
- 1.7 1.7 Statistical Channel Models . . . 6
- 1.8 1.8 Probability, Random Variables, and Random Processes . . . 6
- 1.9 1.9 Fourier Transform Reference . . . 7

2 Section 2 — Practice Exam: Full Worked Solutions **9**
- 2.1 Problem 1 (30 pts): Finding $x_I(t)$, $x_Q(t)$ from $x_c(t)$ . . . 9
- 2.2 Problem 2 (40 pts): Passband Noise through I/Q Downconverter . . . 9
- 2.3 Problem 3 (60 pts): Two-Ray Fading Channel Model . . . 11
  - 2.3.1 Part 1(a): Delays τ_1 and τ_2 . . . 12
  - 2.3.2 Part 1(b): Channel Impulse Response $h(t)$ . . . 12
  - 2.3.3 Part 1(c): Transfer Function $\tilde{h}(f)$ . . . 12
  - 2.3.4 Part 2(a): Delay Spread, Doppler Frequencies . . . 13
  - 2.3.5 Part 2(b): Time-Varying $h(t)$ under Flat-Fading Approximation . . . 13
- 2.4 Additional Topics Not Covered by Practice Exam . . . 13

3 Section 3 — Key Concepts & Definitions **16**
- 3.1 3.1 I/Q Up and Down Converter Layer . . . 16
- 3.2 3.2 Random Processes and Noise . . . 16
- 3.3 3.3 Wireless Channel Models . . . 16
- 3.4 3.4 Probability and Statistics Reference . . . 17
- 3.5 3.5 Important Notation Summary . . . 18

Section 1 — Equation Reference Sheet

1.1 Passband / Baseband Signal Representation

Passband–Baseband Relation

$$x_c(t) = \mathfrak{Re}\left\{\sqrt{2}\,x(t)\,e^{j2\pi f_c t}\right\}, \qquad x(t) = x_I(t) + jx_Q(t) \tag{1}$$

What it means: Every real passband signal $x_c(t)$ at carrier f_c can be written in terms of its *complex baseband envelope* $x(t)$. x_I is the in-phase component; x_Q is the quadrature component.
Use: Converts passband $\leftrightarrow$ baseband (Problem 1 of practice exam).

Complex Envelope Extraction

$$x(t) = \frac{\sqrt{2}\,x_c(t)\,e^{-j2\pi f_c t} + \text{(filtered)}}{} \implies x(t) = \frac{1}{\sqrt{2}}\left[x_c(t) + j\hat{x}_c(t)\right]e^{-j2\pi f_c t} \tag{2}$$

where $\hat{x}_c(t)$ is the Hilbert transform of $x_c(t)$.
Use: Finding $x(t)$ from a given passband signal; used in the I/Q down-converter.

Frequency Domain Passband–Baseband

$$\tilde{x}_c(f) = \frac{1}{\sqrt{2}}[\tilde{x}(f - f_c) + \tilde{x}^*(-f - f_c)] \tag{3}$$

$$\tilde{x}(f) = \sqrt{2}\,\tilde{x}_c(f + f_c)\,\tilde{h}_L(f), \quad \tilde{h}_L(f) = \text{rect}_b(f) \tag{4}$$

What it means: The passband spectrum is the baseband spectrum shifted to $\pm f_c$ (scaled by $1/\sqrt{2}$); to recover baseband, shift back and low-pass filter.
Use: Computing $\tilde{x}_c(f)$ from $\tilde{x}(f)$ and vice versa.

In-Phase & Quadrature from Complex Exponential

$$\cos(\theta) = \mathfrak{Re}\left\{e^{j\theta}\right\}, \qquad \sin(\theta) = \mathfrak{Re}\left\{-je^{j\theta}\right\} \tag{5}$$

Use: Converting cos / sin passband expressions to find x_I, x_Q (standard algebraic trick used throughout).

Amplitude and Phase Envelope

$$x(t) = x_A(t)\,e^{jx_P(t)}, \quad x_A(t) = |x(t)|, \quad x_P(t) = \angle x(t) \tag{6}$$

$x_A(t)$ is the *amplitude envelope*; $x_P(t)$ is the *instantaneous phase*.

1.2 I/Q Up and Down Converter

I/Q Up Converter (Transmitter)

$$x_c(t) = \sqrt{2}\,x_I(t)\cos(2\pi f_c t) - \sqrt{2}\,x_Q(t)\sin(2\pi f_c t) \tag{7}$$

What it means: Modulates the in-phase component onto a cosine carrier and the quadrature component onto a sine carrier (90° phase shifted).
Use: Transmitter design; verifying passband signals.

I/Q Down Converter (Receiver)

$$x_I(t) = \sqrt{2}\,\text{LPF}\{x_c(t)\cos(2\pi f_c t)\} \tag{8}$$

$$x_Q(t) = -\sqrt{2}\,\text{LPF}\{x_c(t)\sin(2\pi f_c t)\} \tag{9}$$

What it means: Mixes the received passband signal with local oscillator signals and low-pass filters to recover the I and Q components.

1.3 Power Spectral Density (PSD) Relationships

PSD of Passband Signal from Baseband PSD

$$\tilde{r}_{x_c}(f) = \frac{\tilde{r}_x(f - f_c) + \tilde{r}_x(-f - f_c)}{2} \tag{10}$$

What it means: The PSD of the passband process is the baseband PSD shifted to $\pm f_c$, averaged.
Use: Converting between baseband and passband noise PSDs (Problem 2).

Baseband PSD from Passband PSD

$$\tilde{r}_n(f) = 2\,\tilde{r}_{n_c}(f + f_c)\,|\tilde{h}_L(f)|^2 \tag{11}$$

What it means: Shift passband PSD down by f_c, multiply by 2, and apply ideal LPF to get complex baseband PSD.
Use: Computing $\tilde{r}_n(f)$ from a given passband filter and white noise input (Problem 2).

PSD of In-Phase Noise Component

$$\tilde{r}_{n_I}(f) = \frac{\tilde{r}_n(f) + \tilde{r}_n(-f)}{4} \tag{12}$$

What it means: The PSD of the in-phase noise equals the *even part* of the baseband PSD, scaled by $\frac{1}{2}$ (factor of $\frac{1}{4}$ because $\frac{1}{2}$ from even part, $\frac{1}{2}$ from $r_{n_I} = \frac{1}{2}\mathrm{Re}\{r_n\}$).
Use: Computing $\tilde{r}_{n_I}(f)$; then integrating to find variance.

Cross-PSD: I and Q Components

$$\tilde{r}_{n_I,n_Q}(f) = j\,\frac{\tilde{r}_n(f) - \tilde{r}_n(-f)}{4} \tag{13}$$

What it means: The cross-PSD equals j times the *odd part* of $\tilde{r}_n(f)$, scaled by $\frac{1}{4}$. If $\tilde{r}_n(f)$ is symmetric (even), then n_I and n_Q are uncorrelated (and independent if Gaussian).
Use: Determining independence of n_I and n_Q (Problem 2, part 4).

PSD of Quadrature Component $n_Q(t)$

$$\tilde{r}_{n_Q}(f) = \frac{\tilde{r}_n(f) + \tilde{r}_n(-f)}{4} = \tilde{r}_{n_I}(f) \tag{14}$$

What it means: The in-phase and quadrature components always have *identical* PSDs (and thus identical variance/power).

Baseband PSD from In-Phase and Quadrature PSDs

$$\tilde{r}_n(f) = 2\tilde{r}_{n_I}(f) - 2j\,\tilde{r}_{n_I,n_Q}(f) \tag{15}$$

Variance from PSD – WSS Process

$$\sigma^2_{n_I} = r_{n_I}(0) = \int_{-\infty}^{\infty} \tilde{r}_{n_I}(f)\,df = 2\int_0^{\infty} \tilde{r}_{n_I}(f)\,df \tag{16}$$

What it means: For a WSS process, variance equals the total area under the PSD (Wiener–Khinchin). The last equality uses the fact that $\tilde{r}_{n_I}(f)$ is *even*.
Use: Computing noise variance (Problem 2, part 5).

1.4 White Noise and Filtering

White Gaussian Noise (WGN) PSD

$$\tilde{r}_w(f) = \frac{N_0}{2} \quad \forall f \tag{17}$$

What it means: The PSD of real-valued WGN is flat at $N_0/2$ for all frequencies (two-sided). The autocorrelation

is $r_w(\tau) = \frac{N_0}{2}\delta(\tau)$.
Use: Starting point for noise PSD calculations (Problem 2).

Output PSD of a Linear Filter

$$\tilde{r}_{n_c}(f) = \tilde{r}_w(f) \cdot |\tilde{h}_c(f)|^2 = \frac{N_0}{2}|\tilde{h}_c(f)|^2 \tag{18}$$

What it means: Filtering multiplies the PSD by $|H(f)|^2$ (power spectral response).
Use: Computing output noise PSD after a passband filter.

Correlation of Filtered WSS Process

$$r_y(\tau) = r_x(\tau) * h(\tau) * h^*(-\tau), \qquad \tilde{r}_y(f) = \tilde{r}_x(f)|H(f)|^2 \tag{19}$$

Use: Finding the PSD of any filtered stationary process.

1.5 Complex Baseband Channel (Wireless Multipath)

General Received Baseband Signal (Multipath)

$$y(t) = \sum_{i=1}^{L} a_i(t)\, e^{-j2\pi f_c \tau_i(t)}\, x(t - \tau_i(t)) \tag{20}$$

where $a_i(t) = \frac{\beta}{r_i(t)} e^{-j\phi_i}$, $r_i(t)$ is path length, ϕ_i is reflection phase.
What it means: The received signal is a superposition of delayed, attenuated, phase-rotated copies of the transmitted baseband signal, one per path.
Use: Two-ray fading model analysis (Problem 3).

Baseband Channel Impulse Response (Static)

$$h(t) = \sum_{i=1}^{L} a_i\, e^{-j2\pi f_c \tau_i}\, \delta(t - \tau_i), \qquad y(t) = h(t) * x(t) \tag{21}$$

What it means: When static (no motion), the channel is LTI with impulse response $h(t)$ consisting of weighted, delayed impulses.
Use: Part 1 of Problem 3.

Baseband Channel Transfer Function (Static)

$$\tilde{h}(f) = \mathcal{F}\{h(t)\} = \sum_{i=1}^{L} a_i\, e^{-j2\pi f_c \tau_i}\, e^{-j2\pi f \tau_i} \tag{22}$$

Use: Computing frequency response of the static multipath channel.

Path Delay and Path Length

$$\tau_i = \frac{r_i}{c}, \qquad c = 3 \times 10^8 \text{ m/s} \tag{23}$$

Two-Ray Model Geometry

$$r_1 = \sqrt{d_{UV}^2 + (d_U - d_V)^2} \quad \text{(LOS path)} \tag{24}$$

$$r_2 = \sqrt{d_{UV}^2 + (d_U + d_V)^2} \quad \text{(reflected path)} \tag{25}$$

Use: Computing path lengths in two-ray model (Problem 3).

1.6 Doppler Effect and Time-Varying Channels

Doppler Frequency for Single Path

$$f_{d,i}(t) = -f_c\,\tau_i'(t) = -f_c\,\frac{r_i'(t)}{c} \tag{26}$$

What it means: The Doppler shift equals $-f_c$ times the time derivative of the path delay (i.e., relative speed divided by c times f_c).
Use: Computing Doppler frequencies (Problem 3, part 2).

Derivative of Path Length (Radial Velocity)

$$r_i'(t) = \frac{v\,d_{UV}(t)}{\sqrt{d_{UV}^2(t) + (d_U \pm d_V)^2}} \tag{27}$$

where $v = \dot{d}_{UV}$ is the relative speed between Tx and Rx.
Use: Computing Doppler for two-ray geometry.

Delay Spread and Doppler Spread

$$\Delta\tau = \max_i \tau_i - \min_i \tau_i = \frac{|r_1 - r_2|}{c} \tag{28}$$

$$\Delta f_d = |f_{d,1} - f_{d,2}| \tag{29}$$

What it means: Delay spread quantifies multipath-induced time dispersion; Doppler spread quantifies frequency dispersion due to motion.

First-Order Taylor: Time-Varying Phase

$$e^{-j2\pi f_c \tau_i(t)} \approx e^{-j2\pi f_c \tau_i(0)}\, e^{j2\pi f_{d,i} t} \tag{30}$$

What it means: For slowly varying channels, the carrier phase factor can be decomposed into a static phase and a Doppler tone.
Use: Narrowband channel model with Doppler (Problem 3, part 2b).

Time-Varying Channel Coefficient (Flat Fading Approximation)

$$h(t) = \sum_{i=1}^{L} a_i\, e^{-j2\pi f_c \tau_i(t)} \approx \sum_{i=1}^{L} a_i\, e^{-j2\pi f_c \tau_i(0)}\, e^{j2\pi f_{d,i} t} \tag{31}$$

Use: Computing $h(t)$ when bandwidth $\gg$ delay spread (Problem 3, part 2b).

Coherence Time and Doppler Spread

$$T_c \approx \frac{1}{\Delta f_d} \tag{32}$$

What it means: The coherence time is approximately the inverse of the Doppler spread. If symbol period $\ll T_c$, the channel is *slow fading*; if symbol period $\gg T_c$, it is *fast fading*.

Coherence Bandwidth and Delay Spread

$$B_c \approx \frac{1}{\Delta\tau} \tag{33}$$

What it means: The coherence bandwidth is approximately the inverse of the delay spread. If signal bandwidth $\ll B_c$, channel is *frequency flat*; if signal bandwidth $\gg B_c$, channel is *frequency selective*.

1.7 Statistical Channel Models

Rayleigh Fading (CSCG Channel Coefficient)

$$h \sim \mathcal{CN}(0, \sigma_h^2), \qquad |h|^2 \sim \mathrm{Exp}\left(\frac{1}{\sigma_h^2}\right) \tag{34}$$

What it means: When many unresolvable paths combine with random phases, the channel coefficient is circularly symmetric complex Gaussian (CSCG), leading to Rayleigh-distributed amplitude.
Use: Flat fading channel model; worst-case statistical model.

Power Delay Profile

$$\sigma_{h_\tau}^2 \triangleq \mathbb{E}\left[|h_i|^2 : \tau_i = \tau\right] \tag{35}$$

The RMS delay spread is $\sigma_\tau = \sqrt{\frac{\sum_i \tau_i^2 \sigma_{h_{\tau_i}}^2}{\sum_i \sigma_{h_{\tau_i}}^2} - \left(\frac{\sum_i \tau_i \sigma_{h_{\tau_i}}^2}{\sum_i \sigma_{h_{\tau_i}}^2}\right)^2}$.
Use: Frequency selective fading channel modeling.

1.8 Probability, Random Variables, and Random Processes

Conditional Probability and Independence

$$P(A|B) = \frac{P(AB)}{P(B)}, \qquad A \perp B \Leftrightarrow P(AB) = P(A)P(B) \tag{36}$$

Gaussian (Normal) PDF — Scalar

$$x \sim \mathcal{N}(\mu_x, \sigma_x^2), \quad f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(-\frac{|x-\mu_x|^2}{2\sigma_x^2}\right) \tag{37}$$

Use: Noise modeling; detection statistics.

Vector Gaussian PDF

$$\vec{x} \sim \mathcal{N}(\vec{\mu}_{\vec{x}}, \Sigma_{\vec{x}}), \quad f_{\vec{x}}(\vec{x}) = \frac{1}{\sqrt{\det(2\pi\Sigma_{\vec{x}})}} \exp\left(-\frac{(\vec{x}-\vec{\mu}_{\vec{x}})^\dagger \Sigma_{\vec{x}}^{-1} (\vec{x}-\vec{\mu}_{\vec{x}})}{2}\right) \tag{38}$$

Use: Multivariate noise; MIMO channel analysis.

CSCG (Circularly Symmetric Complex Gaussian) PDF

$$z \sim \mathcal{CN}(0, \sigma^2), \quad f_z(z) = \frac{1}{\pi\sigma^2} \exp\left(-\frac{|z|^2}{\sigma^2}\right) \tag{39}$$

Properties: $z = z_I + jz_Q$, where $z_I, z_Q \sim \mathcal{N}(0, \sigma^2/2)$ are *independent*; $\mathbb{E}[z^2] = 0$; $\mathbb{E}[|z|^2] = \sigma^2$.
Use: Complex noise; Rayleigh fading channel coefficient.

CSCG Vector PDF

$$\vec{z} \sim \mathcal{CN}(0, \Sigma_{\vec{z}}), \quad f_{\vec{z}}(\vec{z}) = \frac{1}{\pi^n \det(\Sigma_{\vec{z}})} \exp\left(-\vec{z}^\dagger \Sigma_{\vec{z}}^{-1} \vec{z}\right) \tag{40}$$

where n is the dimension of $\vec{z}$.

Mean – Variance – Correlation

$$\mu_x = \mathbb{E}[x] = \int x\, f_x(x)\, dx \tag{41}$$

$$\sigma_x^2 = \mathbb{E}[|x-\mu_x|^2] \tag{42}$$

$$\mathbb{E}[xy^*] = \text{correlation of } x \text{ and } y \tag{43}$$

$$\mathbb{E}[(x-\mu_x)(y-\mu_y)^*] = \text{covariance of } x \text{ and } y \tag{44}$$

Wide-Sense Stationary (WSS) Process

$$\mu_x(t) = \mathbb{E}[x(t)] = \mu_x \quad \text{(constant)} \tag{45}$$

$$r_x(t+\tau, t) = \mathbb{E}[x(t+\tau)x^*(t)] = r_x(\tau) \quad \text{(depends only on } \tau\text{)} \tag{46}$$

What it means: Mean is constant and autocorrelation depends only on lag τ, not absolute time.
Key implication: Variance is time-independent: $\sigma_x^2 = r_x(0)$.

Wiener–Khinchin Theorem

$$\tilde{r}_x(f) = \mathcal{F}\{r_x(\tau)\}, \qquad r_x(\tau) = \mathcal{F}^{-1}\{\tilde{r}_x(f)\} \tag{47}$$

What it means: The PSD of a WSS process is the Fourier transform of its autocorrelation function.
Use: Connecting time-domain correlations to frequency-domain PSDs.

AWGN Properties

A zero-mean, white, real, stationary, Gaussian noise process $w(t)$:

$$r_w(\tau) = \frac{N_0}{2}\delta(\tau) \quad \text{(real-valued)} \tag{48}$$

$$\tilde{r}_w(f) = \frac{N_0}{2} \quad \forall f \tag{49}$$

$$w(t_1),\ w(t_2) \text{ independent for } t_1 \neq t_2 \tag{50}$$

Circularly Symmetric WSS Complex Gaussian Process

A complex process $x(t) = x_I(t) + jx_Q(t)$ is CSCG and WSS if:

- $\mathbb{E}[x(t)] = 0$
- $\mathbb{E}[x(t+\tau)x^*(t)] = r_x(\tau)$ (only depends on τ)
- $\mathbb{E}[x(t+\tau)x(t)] = 0$ (circularity condition)

Then x_I, x_Q are jointly Gaussian, each has PSD $\tilde{r}_{n_I}(f) = \frac{\tilde{r}_n(f)+\tilde{r}_n(-f)}{4}$.

Complementary Error Function

$$\frac{1}{2}\operatorname{erfc}(x_0) = P\{x > x_0\}, \quad x \sim \mathcal{N}\left(0, \tfrac{1}{2}\right) \tag{51}$$

For $x \sim \mathcal{N}(0,1)$: $P\{x > x_0\} = \frac{1}{2}\operatorname{erfc}\left(\frac{x_0}{\sqrt{2}}\right) = Q(x_0)$, where $Q(x_0) = \frac{1}{\sqrt{2\pi}}\int_{x_0}^{\infty} e^{-u^2/2}\,du$.
Use: Bit error probability calculations.

Markov Inequality

$$P\{x \geq a\} \leq \frac{\mathbb{E}[x]}{a}, \quad x \geq 0,\ a > 0 \tag{52}$$

1.9 Fourier Transform Reference

Key Fourier Transform Pairs

$$\mathcal{F}\{\delta(t)\} = 1 \tag{53}$$

$$\mathcal{F}\{e^{j2\pi f_0 t}\} = \delta(f - f_0) \tag{54}$$

$$\mathcal{F}\{\cos(2\pi f_0 t)\} = \frac{\delta(f - f_0) + \delta(f + f_0)}{2} \tag{55}$$

$$\mathcal{F}\{\sin(2\pi f_0 t)\} = \frac{\delta(f - f_0) - \delta(f + f_0)}{2j} \tag{56}$$

$$\mathcal{F}\{\mathrm{rect}_a(t)\} = a\,\mathrm{sinc}(fa), \quad \mathrm{rect}_a(t) = 1 \text{ if } |t| \leq a/2 \tag{57}$$

$$\mathcal{F}\{x(t - \tau_0)\} = e^{-j2\pi f \tau_0}\,\tilde{x}(f) \tag{58}$$

$$\mathcal{F}\{x(t)e^{j2\pi f_0 t}\} = \tilde{x}(f - f_0) \tag{59}$$

$$\mathcal{F}\{x(t) * y(t)\} = \tilde{x}(f)\,\tilde{y}(f) \tag{60}$$

$$\mathcal{F}\{x(t)y(t)\} = \tilde{x}(f) * \tilde{y}(f) \tag{61}$$

Euler's Formula

$$e^{j\theta} = \cos\theta + j\sin\theta, \quad \cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \quad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j} \tag{62}$$

Section 2 — Practice Exam: Full Worked Solutions

Problem 1 (30 pts): Finding $x_I(t)$, $x_Q(t)$ from $x_c(t)$

Problem Statement

Find $x_I(t)$, $x_Q(t)$ of the baseband signal $x(t)$ that produces the following passband signal $x_c(t)$.
1. (5/10 pts) $x_c(t) = \sqrt{8}\, m(t) \cos\big(2\pi f_c t + \frac{\pi}{4}\big)$, where $m(t)$ is a real-valued baseband message signal.
2. (5/10 pts) $x_c(t) = 4\cos(2\pi(f_c + f_\Delta)t)$, where f_Δ is a constant frequency offset ($f_\Delta \ll f_c$).

Solution Strategy: Use the passband–baseband relation $x_c(t) = \mathfrak{Re}\big\{\sqrt{2}\, x(t)\, e^{j2\pi f_c t}\big\}$. Rewrite $x_c(t)$ in the canonical form $x_c(t) = \mathfrak{Re}\big\{C\, e^{j2\pi f_c t}\big\}$, equate $C = \sqrt{2}\, x(t)$, then read off $x_I = \mathfrak{Re}\{x\}$ and $x_Q = \mathfrak{Im}\{x\}$.
Relevant Equations:

$$x_c(t) = \mathfrak{Re}\Big\{\sqrt{2}\, x(t)\, e^{j2\pi f_c t}\Big\}, \quad \cos\theta = \mathfrak{Re}\{e^{j\theta}\}, \quad e^{j\pi/4} = \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$$

Part 1 — Step-by-Step:
Step 1: Convert cosine to complex exponential.

$$x_c(t) = \sqrt{8}\, m(t)\, \mathfrak{Re}\Big[e^{j(2\pi f_c t + \pi/4)}\Big] = \mathfrak{Re}\Big[\sqrt{8}\, m(t)\, e^{j\pi/4} e^{j2\pi f_c t}\Big]$$

(Since $m(t)$ is real, it can be pulled inside $\mathfrak{Re}\{\cdot\}$.)
Step 2: Match to standard form $\mathfrak{Re}\big[\sqrt{2}\, x(t)\, e^{j2\pi f_c t}\big]$.

$$\sqrt{2}\, x(t) = \sqrt{8}\, m(t)\, e^{j\pi/4} \implies x(t) = \frac{\sqrt{8}}{\sqrt{2}}\, m(t)\, e^{j\pi/4} = 2m(t)\, e^{j\pi/4}$$

Step 3: Expand using Euler's formula.

$$x(t) = 2m(t)\left(\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}\right) = \sqrt{2}\, m(t) + j\sqrt{2}\, m(t)$$

Part 1 Final Answer

$$x_I(t) = \sqrt{2}\, m(t), \qquad x_Q(t) = \sqrt{2}\, m(t)$$

Part 2 — Step-by-Step:
Step 1: Expand $x_c(t) = 4\cos(2\pi f_c t + 2\pi f_\Delta t)$ and convert.

$$x_c(t) = \mathfrak{Re}\Big[4e^{j2\pi(f_c + f_\Delta)t}\Big] = \mathfrak{Re}\big[4e^{j2\pi f_\Delta t}\, e^{j2\pi f_c t}\big]$$

Step 2: Match to standard form.

$$\sqrt{2}\, x(t) = 4e^{j2\pi f_\Delta t} \implies x(t) = \frac{4}{\sqrt{2}} e^{j2\pi f_\Delta t} = 2\sqrt{2}\, e^{j2\pi f_\Delta t}$$

Step 3: Expand.

$$x(t) = 2\sqrt{2}\big[\cos(2\pi f_\Delta t) + j\sin(2\pi f_\Delta t)\big]$$

Part 2 Final Answer

$$x_I(t) = 2\sqrt{2}\cos(2\pi f_\Delta t), \qquad x_Q(t) = 2\sqrt{2}\sin(2\pi f_\Delta t)$$

Note: This is a frequency offset signal — the baseband signal is a complex exponential at f_Δ.

Problem 2 (40 pts): Passband Noise through I/Q Downconverter

Problem Statement

Zero-mean white real stationary Gaussian noise $w(t)$ with $r_w(\tau) = \frac{N_0}{2}\delta(\tau)$ is input to passband filter $\tilde{h}_c(f)$ with output $n_c(t)$, then to an I/Q down-converter producing $n(t) = n_I(t) + jn_Q(t)$. Given $f_c = 1000$ Hz and asymmetric

filter:

$$\tilde{h}_c(f) = \begin{cases} \sqrt{\frac{q}{2}} \cdot \sqrt{2} & 500 \le |f| \le 1000 \\ 4 - \frac{|f|}{500} & 1000 < |f| \le 2000 \\ 0 & \text{elsewhere} \end{cases}$$

Find: (1) $\tilde{r}_{n_c}(f)$, (2) $\tilde{r}_n(f)$, (3) $\tilde{r}_{n_I}(f)$, (4) $\tilde{r}_{n_I,n_Q}(f)$ and independence of n_I, n_Q, (5) $\sigma^2_{n_I}(3)$.

Solution Strategy: (1) Multiply WGN PSD by $|\tilde{h}_c(f)|^2$. (2) Shift PSD down by $f_c = 1000$ Hz and multiply by 2. (3) Take even part of $\tilde{r}_n(f)$ and divide by 4. (4) Take j times odd part divided by 4; check if zero. (5) Integrate $\tilde{r}_{n_I}(f)$ over all f.

Relevant Equations:

$$\tilde{r}_{n_c}(f) = \frac{N_0}{2}|\tilde{h}_c(f)|^2, \quad \tilde{r}_n(f) = 2\tilde{r}_{n_c}(f + f_c), \quad \tilde{r}_{n_I}(f) = \frac{\tilde{r}_n(f) + \tilde{r}_n(-f)}{4}, \quad \tilde{r}_{n_I,n_Q}(f) = j\frac{\tilde{r}_n(f) - \tilde{r}_n(-f)}{4}$$

Part 1 — Compute $|\tilde{h}_c(f)|^2$:

For $500 \le |f| \le 1000$: $|\tilde{h}_c(f)|^2 = 2$, so $\tilde{r}_{n_c} = \frac{N_0}{2} \cdot 2 = N_0$.

For $1000 < |f| \le 2000$: $|\tilde{h}_c(f)|^2 = \left(4 - \frac{|f|}{500}\right)^2$. At $|f| = 1000$: $(4-2)^2 = 4$, so $\tilde{r}_{n_c} = 2N_0$. Hmm, but the given solution uses $|\tilde{h}_c(f)|^2$ directly.

Per the solution, interpreting $\tilde{h}_c(f)$ as given (with $\sqrt{q/2} \cdot \sqrt{2}$ likely implying $|\tilde{h}_c|^2 = 2$ for lower band):

Part 1: PSD of Passband Noise

$$\tilde{r}_{n_c}(f) = \begin{cases} N_0 & 500 \le |f| \le 1000 \\ N_0\left(2 - \frac{|f|}{1000}\right)^2 & \text{(simplified form per professor's solution)} \\ 0 & \text{elsewhere} \end{cases}$$

Per professor's solution:

$$\tilde{r}_{n_c}(f) = \begin{cases} N_0 & 500 \le |f| \le 1000 \\ N_0\left(2 - \frac{|f|}{1000}\right) & 1000 < |f| \le 2000 \\ 0 & \text{elsewhere} \end{cases}$$

Part 2 — Compute $\tilde{r}_n(f)$:

Shift $\tilde{r}_{n_c}(f)$ down by $f_c = 1000$ Hz (i.e., replace $f \to f + f_c$ in the passband PSD) and multiply by 2:

$$\tilde{r}_n(f) = 2\tilde{r}_{n_c}(f + 1000)$$

At $f + 1000 = 1000 \Rightarrow f = 0$; at $f + 1000 = 500 \Rightarrow f = -500$; at $f + 1000 = 2000 \Rightarrow f = 1000$.

Part 2: Complex Baseband PSD

$$\tilde{r}_n(f) = \begin{cases} 2N_0 & -500 \le f \le 0 \\ 2N_0\left(1 - \frac{f}{1000}\right) & 0 < f \le 1000 \\ 0 & \text{elsewhere} \end{cases}$$

Note: $\tilde{r}_n(f)$ is **asymmetric** (not even) because the passband filter was asymmetric around f_c.

Part 3 — Compute $\tilde{r}_{n_I}(f)$:

Use $\tilde{r}_{n_I}(f) = \frac{\tilde{r}_n(f) + \tilde{r}_n(-f)}{4}$.

We need $\tilde{r}_n(-f)$:

$$\tilde{r}_n(-f) = \begin{cases} 2N_0 & 0 \le f \le 500 \\ 2N_0\left(1 + \frac{f}{1000}\right) & -1000 \le f < 0 \Rightarrow \tilde{r}_n(-f) = 2N_0\left(1 + \frac{|f|}{1000}\right) \text{ for } 0 \le f \le 1000 \\ 0 & \text{elsewhere} \end{cases}$$

Wait, let me be more careful. $\tilde{r}_n(-f)$: replace $f \to -f$ in $\tilde{r}_n(f)$:

$$\tilde{r}_n(-f) = \begin{cases} 2N_0 & 0 \le f \le 500 \\ 2N_0\left(1 + \frac{f}{1000}\right) & -1000 \le f < -500 \Rightarrow 500 < f \le 1000 \\ 0 & \text{elsewhere} \end{cases}$$

Part 3: In-Phase PSD

$$\tilde{r}_{n_I}(f) = \begin{cases} \frac{N_0}{2}\left(2 - \frac{|f|}{1000}\right) & |f| \le 500 \\ \frac{N_0}{2}\left(1 - \frac{|f|}{1000}\right) & 500 < |f| \le 1000 \\ 0 & \text{elsewhere} \end{cases}$$

$\tilde{r}_{n_I}(f)$ is even (as expected for a PSD).

Part 4 — Compute $\tilde{r}_{n_I,n_Q}(f)$ and independence:

$$\tilde{r}_{n_I,n_Q}(f) = j\,\frac{\tilde{r}_n(f) - \tilde{r}_n(-f)}{4}$$

For $0 < f \le 500$: $\tilde{r}_n(f) = 2N_0\left(1 - \frac{f}{1000}\right)$, $\tilde{r}_n(-f) = 2N_0$. Difference: $2N_0\left(1 - \frac{f}{1000}\right) - 2N_0 = -\frac{2N_0 f}{1000}$.

Part 4: Cross-PSD and Independence

$$\tilde{r}_{n_I,n_Q}(f) = \begin{cases} -j\frac{N_0 f}{2000} & 0 < f \le 500 \\ j\frac{N_0}{2}\left(1 - \frac{f}{1000}\right) & 500 < f \le 1000 \\ \text{(Hermitian symmetry for } f < 0) \end{cases}$$

Since $\tilde{r}_{n_I,n_Q}(f) \neq 0$, $n_I(t)$ and $n_Q(t)$ are **not uncorrelated** $\Rightarrow$ **not independent**.
Physical reason: The passband filter is asymmetric around f_c, so the baseband PSD $\tilde{r}_n(f)$ is not even.

Part 5 — Compute $\sigma^2_{n_I}(3)$:
Since $n_I(t)$ is WSS, the variance is *time-invariant*: $\sigma^2_{n_I}(3) = \sigma^2_{n_I} = r_{n_I}(0) = \int_{-\infty}^{\infty} \tilde{r}_{n_I}(f)\,df$.
Using the even symmetry: $\sigma^2_{n_I} = 2\int_0^{1000} \tilde{r}_{n_I}(f)\,df$.
Integral over $[0, 500]$:

$$\int_0^{500} \frac{N_0}{2}\left(2 - \frac{f}{1000}\right) df = \frac{N_0}{2}\left[2f - \frac{f^2}{2000}\right]_0^{500} = \frac{N_0}{2}\left(1000 - \frac{250000}{2000}\right) = \frac{N_0}{2}(1000 - 125) = 437.5\,N_0$$

Wait, let me redo: $\frac{N_0}{2} \cdot 875 = 437.5N_0$.
Integral over $[500, 1000]$:

$$\int_{500}^{1000} \frac{N_0}{2}\left(1 - \frac{f}{1000}\right) df = \frac{N_0}{2}\left[f - \frac{f^2}{2000}\right]_{500}^{1000} = \frac{N_0}{2}[(1000 - 500) - (500 - 125)] = \frac{N_0}{2}(500 - 375) = 62.5\,N_0$$

Total: $\sigma^2_{n_I} = 2(437.5N_0 + 62.5N_0) = 2 \times 500N_0 = 1000N_0$.

Part 5 Final Answer

$$\boxed{\sigma^2_{n_I}(3) = 1000\,N_0}$$

Key insight: The answer is time-invariant (does not depend on $t = 3$) because $n_I(t)$ is WSS.

Problem 3 (60 pts): Two-Ray Fading Channel Model

Problem Statement

A UAV transmits to a ground vehicle. Path 1 is LOS; Path 2 is ground-reflected (perfect conductor, adds phase π).

Received baseband signal: $y(t) = \sum_{i=1}^{2} a_i e^{-j2\pi f_c \tau_i(t)} x(t - \tau_i(t))$, where $a_i = \frac{\beta}{r_i(t)} e^{-j\phi_i}$, $\phi_1 = 0$, $\phi_2 = \pi$.

$$r_1(t) = \sqrt{d_{UV}^2 + (d_U - d_V)^2}, \quad r_2(t) = \sqrt{d_{UV}^2 + (d_U + d_V)^2}$$

Parameters: $f_c = 2.4$ GHz, $d_U = 50$ m, $d_V = 2$ m, $d_{UV} = d_0 + vt$, $d_0 = 200$ m.
Part 1 ($v = 0$): (a) Find τ_1, τ_2. (b) Find $h(t)$. (c) Find $\tilde{h}(f)$.
Part 2 ($v = 30$ m/s): (a) Find delay spread $\Delta\tau$, Doppler freq. $f_{d,1}, f_{d,2}$, Doppler spread Δf_d at $t = 0$. (b) Assuming BW $\gg$ delay spread, find $h(t)$.

Solution Strategy: Part 1: With $v = 0$, channel is static $\Rightarrow$ LTI. Compute path lengths, then delays $\tau_i = r_i/c$. Express $h(t)$ as sum of weighted impulses; $\tilde{h}(f)$ is Fourier transform. Part 2: With motion, compute $r_i'(0)$ from derivative of path length. Doppler $f_{d,i} = -f_c r_i'(0)/c$. For flat-fading approximation, $h(t)$ is a time-varying complex coefficient.
Relevant Equations:

$$h(t) = \sum_i a_i e^{-j2\pi f_c \tau_i} \delta(t - \tau_i), \quad \tilde{h}(f) = \sum_i a_i e^{-j2\pi f_c \tau_i} e^{-j2\pi f \tau_i}, \quad f_{d,i} = -f_c \frac{r_i'(t)}{c}$$

Part 1(a): Delays τ_1 and τ_2
With $v = 0$: $d_{UV} = d_0 = 200$ m.

$$r_1 = \sqrt{200^2 + (50-2)^2} = \sqrt{40000 + 2304} = \sqrt{42304} \approx 205.679 \text{ m}$$
$$r_2 = \sqrt{200^2 + (50+2)^2} = \sqrt{40000 + 2704} = \sqrt{42704} \approx 206.649 \text{ m}$$

$$\tau_1 = \frac{r_1}{c} = \frac{205.679}{3 \times 10^8} \approx 6.8560 \times 10^{-7} \text{ s}$$
$$\tau_2 = \frac{r_2}{c} = \frac{206.649}{3 \times 10^8} \approx 6.8883 \times 10^{-7} \text{ s}$$

Part 1(a)

$\tau_1 \approx 685.6$ ns, $\tau_2 \approx 688.8$ ns.

Part 1(b): Channel Impulse Response $h(t)$

$$y(t) = \sum_{i=1}^{2} a_i e^{-j2\pi f_c \tau_i} x(t - \tau_i) = \left(\sum_{i=1}^{2} a_i e^{-j2\pi f_c \tau_i} \delta(t - \tau_i) \right) * x(t)$$

$$h(t) = \frac{\beta}{r_1} e^{-j2\pi f_c \tau_1} \delta(t - \tau_1) + \frac{\beta}{r_2} e^{j\pi} e^{-j2\pi f_c \tau_2} \delta(t - \tau_2)$$

Note: $e^{j\pi} = -1$ (reflection phase for perfect conductor).
Numerically: $f_c \tau_1 = 2.4 \times 10^9 \times 6.856 \times 10^{-7} = 1645.44$ cycles; $f_c \tau_2 = 2.4 \times 10^9 \times 6.888 \times 10^{-7} = 1653.16$ cycles.

Part 1(b)

$$h(t) = \frac{\beta}{205.679} e^{-j2\pi(1645.44)} \delta(t - \tau_1) - \frac{\beta}{206.649} e^{-j2\pi(1653.16)} \delta(t - \tau_2)$$

(The minus sign comes from $e^{j\pi} = -1$ for the reflected path.)

Part 1(c): Transfer Function $\tilde{h}(f)$
$\tilde{h}(f) = \mathcal{F}\{h(t)\}$. Using $\mathcal{F}\{\delta(t - \tau)\} = e^{-j2\pi f \tau}$:

Part 1(c)

$$\tilde{h}(f) = \frac{\beta}{r_1} e^{-j2\pi f_c \tau_1} e^{-j2\pi f \tau_1} - \frac{\beta}{r_2} e^{-j2\pi f_c \tau_2} e^{-j2\pi f \tau_2}$$

Factor out $e^{-j2\pi f\tau_1}$:

$$\tilde{h}(f) = e^{-j2\pi f\tau_1}\left[\frac{\beta}{r_1}e^{-j2\pi f_c\tau_1} - \frac{\beta}{r_2}e^{-j2\pi f_c\tau_2}\, e^{-j2\pi f(\tau_2-\tau_1)}\right]$$

Part 2(a): Delay Spread, Doppler Frequencies

$$\Delta\tau = |\tau_2(0) - \tau_1(0)| = \frac{r_2(0) - r_1(0)}{c} = \frac{206.649 - 205.679}{3\times 10^8} = \frac{0.970}{3\times 10^8} \approx 3.23\times 10^{-9}\ \text{s}$$

For Doppler: $r_i'(t) = \frac{d}{dt}\sqrt{d_{UV}^2(t) + (d_U \pm d_V)^2} = \frac{v\, d_{UV}(t)}{\sqrt{d_{UV}^2(t)+(d_U\pm d_V)^2}}$ (since d_U, d_V are constant, only $d_{UV} = d_0 + vt$ varies).

At $t = 0$: $d_{UV} = 200$ m.

$$r_1'(0) = \frac{30\times 200}{\sqrt{200^2+48^2}} = \frac{6000}{205.679} \approx 29.170\ \text{m/s}$$

$$r_2'(0) = \frac{30\times 200}{\sqrt{200^2+52^2}} = \frac{6000}{206.649} \approx 29.033\ \text{m/s}$$

$$f_{d,1} = -f_c\frac{r_1'(0)}{c} = -2.4\times 10^9 \times \frac{29.170}{3\times 10^8} \approx -233.37\ \text{Hz}$$

$$f_{d,2} = -f_c\frac{r_2'(0)}{c} = -2.4\times 10^9 \times \frac{29.033}{3\times 10^8} \approx -232.27\ \text{Hz}$$

Part 2(a)

$\Delta\tau \approx 3.23$ ns, $f_{d,1} \approx -233.37$ Hz, $f_{d,2} \approx -232.27$ Hz, $\Delta f_d = |f_{d,1} - f_{d,2}| \approx 1.10$ Hz.

Note: Both Dopplers are negative because the UAV and vehicle are moving *apart* (increasing d_{UV}) $\Rightarrow$ the channel introduces a red-shift.

Part 2(b): Time-Varying $h(t)$ under Flat-Fading Approximation

When BW $\gg \Delta\tau$ (i.e., symbol period $\ll$ delay spread): $x(t-\tau_i(t)) \approx x(t-\tau_1(0))$ for all i.

$$y(t) \approx \left(\sum_{i=1}^{2} a_i e^{-j2\pi f_c\tau_i(t)}\right) x(t-\tau_1(0)) = h(t)\cdot x(t-\tau_1(0))$$

Using first-order Taylor: $e^{-j2\pi f_c\tau_i(t)} \approx e^{-j2\pi f_c\tau_i(0)}e^{j2\pi f_{d,i}t}$:

Part 2(b)

$$h(t) = \frac{\beta}{r_1}e^{-j2\pi f_c\tau_1(0)}e^{j2\pi f_{d,1}t} - \frac{\beta}{r_2}e^{-j2\pi f_c\tau_2(0)}e^{j2\pi f_{d,2}t}$$

$$= \frac{\beta}{205.679}e^{-j2\pi(1645.44)}e^{-j2\pi(233.37)t} - \frac{\beta}{206.649}e^{-j2\pi(1653.16)}e^{-j2\pi(232.27)t}$$

$h(t)$ is a **time-varying** complex scalar (not an impulse response), oscillating at the difference Doppler frequency $\Delta f_d \approx 1.10$ Hz.

Additional Topics Not Covered by Practice Exam

The following are key topics from pp. 120–575 that were not tested on the practice exam but are fair game for the midterm.

1. Imperfections in I/Q Converters

I/Q imbalance and **carrier phase offset**: In real implementations, the cosine and sine LO signals may not be perfectly 90° apart or equal in amplitude. This introduces cross-talk between x_I and x_Q. Typically modeled as: $x_{c,\text{out}}(t) = \sqrt{2}(c_I x_I(t)\cos(2\pi f_c t) - c_Q x_Q(t)\sin(2\pi f_c t))$, where c_I, c_Q may differ. **Non-linearity** of amplifiers also distorts the baseband signal.

2. Equivalent Baseband Filter

The equivalent baseband representation of a passband filter $\tilde{h}_c(f)$ is:

$$\tilde{h}(f) = \sqrt{2}\,\tilde{h}_c(f + f_c)\,\tilde{h}_L(f)$$

Used to analyze the end-to-end baseband channel including the passband filter.

3. CSCG Processes: Key Properties

If $x(t) \sim \mathcal{CSCG}$, then: (a) $\mathfrak{Re}\{x(t)\}$ and $\mathfrak{Im}\{x(t)\}$ have equal variances; (b) They are uncorrelated and jointly Gaussian $\Rightarrow$ *independent*; (c) $\mathbb{E}[x(t)^2] = 0$ (pseudo-covariance is zero). Filtering a CSCG process with a linear filter produces another CSCG process.

4. Passband AWGN Channel Model

For passband AWGN: $y_c(t) = x_c(t) + n_c(t)$. In baseband: $y(t) = x(t) + n(t)$, where $n(t)$ is complex baseband noise with PSD $\tilde{r}_n(f) = N_0 \cdot \tilde{h}_L(f)$ (flat within bandwidth). $N_0/2$ is the one-sided noise spectral density.

5. Multiple-Path Channel — General Frequency-Selective Case

General baseband channel model (time-invariant, frequency-selective):

$$h(t) = \sum_{i=1}^{L} \alpha_i\,\delta(t - \tau_i), \qquad \tilde{h}(f) = \sum_{i=1}^{L} \alpha_i\,e^{-j2\pi f \tau_i}$$

where $\alpha_i = \frac{\beta}{r_i} e^{-j2\pi f_c \tau_i}$. Frequency-selective fading occurs when $BW > B_c \approx 1/\Delta\tau$.

6. Flat Fading vs. Frequency Selective Fading

Condition	**Signal BW** $\ll B_c$	**Signal BW** $\gg B_c$
Channel type	Frequency Flat	Frequency Selective
$h(t)$	Scalar complex gain	Multipath impulse response
Effect on signal	Uniform amplitude change	ISI (inter-symbol interference)

7. Slow Fading vs. Fast Fading

Condition	**Symbol period** $\ll T_c$	**Symbol period** $\gg T_c$
Fading type	Slow (quasi-static)	Fast
Channel during symbol	Approximately constant	Changes within symbol

$T_c \approx 1/\Delta f_d$ (coherence time).

8. Rayleigh Fading Statistical Model

Under the "worst case" assumption (many unresolvable paths, random phases), the flat-fading channel coefficient $h \sim \mathcal{CN}(0, \sigma_h^2)$:

$$|h| \sim \text{Rayleigh}(\sigma), \quad f_{|h|}(r) = \frac{2r}{\sigma_h^2} e^{-r^2/\sigma_h^2}, \quad r \geq 0$$

$|h|^2 \sim \text{Exp}(1/\sigma_h^2)$. If a strong LOS component exists: Rician fading.

9. MIMO Channel Model

For N_T transmit and N_R receive antennas:

$$\mathbf{y}(t) = \mathbf{H}(t)\mathbf{x}(t) + \mathbf{n}(t)$$

where $\mathbf{H}(t)$ is the $N_R \times N_T$ channel matrix. Each entry follows flat-fading model if frequency-flat.

10. Sampling and Nyquist Theorem

A baseband signal of bandwidth b must be sampled at $f_s > 2b$ (Nyquist rate) to allow perfect reconstruction. Aliasing occurs if $f_s < 2b$. For passband signals, minimum sampling rate can be lower than $2f_c$ if $b \ll f_c$.

Section 3 — Key Concepts & Definitions

3.1 I/Q Up and Down Converter Layer

Complex Baseband Signal

For a real passband signal $x_c(t)$ centered at f_c, the **complex baseband (envelope) signal** $x(t)$ satisfies $x_c(t) = \mathfrak{Re}\{\sqrt{2}\,x(t)e^{j2\pi f_c t}\}$. The baseband signal $x(t)$ contains the same information as $x_c(t)$ but is centered at DC (zero frequency). It has half the bandwidth.

In-Phase and Quadrature Components

$x_I(t) = \mathfrak{Re}\{x(t)\}$ is the **in-phase (I)** component; $x_Q(t) = \mathfrak{Im}\{x(t)\}$ is the **quadrature (Q)** component. Together they completely describe the passband signal. The factor of $\sqrt{2}$ in the relation is a power normalization convention.

Why Use Baseband Processing?

(1) Sampling rate reduction: baseband bandwidth $b \ll f_c$, so $f_s = 2b$ vs. $f_s = 2f_c$. (2) Digital signal processing can be performed at low rates. (3) The I/Q structure captures all modulation information.

Equivalent Baseband Filter

A passband filter $\tilde{H}_c(f)$ has an equivalent baseband representation $\tilde{H}(f) = \sqrt{2}\,\tilde{H}_c(f + f_c)\tilde{H}_L(f)$, where $\tilde{H}_L(f) = \text{rect}_b(f)$ is an ideal LPF of bandwidth b.

3.2 Random Processes and Noise

Stationary vs. Wide-Sense Stationary

A process is **strictly stationary** (SS) if all joint distributions are time-shift invariant. It is **wide-sense stationary** (WSS) if only the mean and autocorrelation are time-shift invariant. WSS is weaker than SS; for Gaussian processes, WSS $\Leftrightarrow$ SS.

Autocorrelation Properties

For WSS $x(t)$: (1) $r_x(0) \geq 0$ (total power); (2) $r_x(\tau) = r_x^*(-\tau)$ (Hermitian); (3) $|r_x(\tau)| \leq r_x(0)$; (4) $\tilde{r}_x(f) \geq 0$ (PSD is non-negative).

Complex Baseband Autocorrelation Relations

$$r_x(\tau) = 2r_{x_I}(\tau) + 2j\,r_{x_I x_Q}(\tau) \quad (\text{real } r_x(\tau) = 2r_{x_I}(\tau) \text{ if CSCG})$$
$$r_{x_I}(\tau) = \mathfrak{Re}\{r_x(\tau)\}/2, \quad r_{x_I x_Q}(\tau) = \mathfrak{Im}\{r_x(\tau)\}/2$$

White Noise

A process is **white** if its PSD is flat (constant) for all frequencies: $\tilde{r}_w(f) = N_0/2$. This implies samples at different times are uncorrelated. White Gaussian noise (WGN) samples are also independent.

AWGN Channel

$y_c(t) = x_c(t) + n_c(t)$ where $n_c(t)$ is real, white, zero-mean, Gaussian noise with PSD $N_0/2$. In baseband: $y(t) = x(t) + n(t)$ where $n(t)$ is CSCG with PSD N_0 within the signal bandwidth.

3.3 Wireless Channel Models

Multipath Propagation

Wireless signals arrive at the receiver via multiple paths (reflections, diffractions, scattering). Each path has its own delay τ_i, attenuation $|\alpha_i|$, and phase shift $\angle\alpha_i$. The superposition of all paths creates: (a) constructive/destructive interference (**fading**), (b) **inter-symbol interference (ISI)** if delay spread $\gg$ symbol period.

Path Attenuation Model

Each path amplitude scales as $a_i = \frac{\beta}{r_i} e^{-j\phi_i}$, where $\frac{1}{r_i}$ is the free-space path loss factor and ϕ_i is the reflection phase (e.g., $\phi = \pi$ for reflection from a perfect conductor).

Doppler Effect

When the transmitter, receiver, or scatterers move, path lengths change over time $\Rightarrow$ time-varying delays $\tau_i(t) \Rightarrow$ time-varying channel. The Doppler frequency $f_{d,i} = -f_c \dot{\tau}_i(t)$ is proportional to the rate of change of path delay. Positive relative velocity (moving apart): $f_d < 0$ (red shift). Negative relative velocity (moving together): $f_d > 0$ (blue shift).

Channel Classification Summary

- **Flat fading:** All signal frequencies experience the same gain/phase. Occurs when $BW \ll B_c = 1/\Delta\tau$. Channel reduces to a single complex scalar.
- **Frequency selective:** Different frequencies experience different gains. Causes ISI.
- **Slow fading:** Channel is quasi-static during a symbol. Occurs when $T_{sym} \ll T_c = 1/\Delta f_d$.
- **Fast fading:** Channel changes within a symbol. Hard to equalize.

Two-Ray Model Summary

For UAV-to-vehicle: LOS path has $r_1 = \sqrt{d_{UV}^2 + (d_U - d_V)^2}$; reflected path has $r_2 = \sqrt{d_{UV}^2 + (d_U + d_V)^2}$. Ground reflection adds π phase. The interference between the two paths creates deep fades when $e^{-j2\pi f_c \tau_1} \approx e^{-j2\pi f_c \tau_2}$ (constructive) or $\approx -e^{-j2\pi f_c \tau_2}$ (destructive, near-null). At low heights, path lengths are very similar and the channel can null completely.

MIMO Channel Key Facts

With N_T transmit, N_R receive antennas: channel matrix $\mathbf{H}$ is $N_R \times N_T$. Capacity scales with $\min(N_T, N_R)$ for independent channels. Spatial multiplexing sends $\min(N_T, N_R)$ independent streams. Spatial diversity sends the same data over multiple antennas to combat fading.

3.4 Probability and Statistics Reference

Gaussian Properties Critical for This Course

- Linear combinations of Gaussian RVs are Gaussian.
- For Gaussian RVs, uncorrelated $\Rightarrow$ independent.
- A filtered Gaussian process is Gaussian.
- $x \sim \mathcal{N}(0, \sigma^2) \Rightarrow |x| \sim$ Rayleigh (from two Gaussians)

CSCG Key Facts

$z \sim \mathcal{CN}(0, \sigma^2)$: (1) $\mathbb{E}[z] = 0$; (2) $\mathbb{E}[|z|^2] = \sigma^2$ (total power); (3) $\mathbb{E}[z^2] = 0$ (circularity); (4) $z = z_I + jz_Q$, $z_I, z_Q \sim \mathcal{N}(0, \sigma^2/2)$, independent; (5) $e^{j\theta} z$ has same distribution as z for any constant θ (rotationally invariant).

Q-Function and BER

$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$. For BPSK over AWGN: $P_e = Q(\sqrt{2E_b/N_0})$. For QPSK: $P_e \approx Q(\sqrt{2E_b/N_0})$ per bit.

Power Spectral Density: Key Facts

(1) $\tilde{r}_x(f) \geq 0$ always; (2) $\tilde{r}_x(f)$ is real and even for real processes; (3) Total power $= r_x(0) = \int \tilde{r}_x(f) df$; (4) For complex processes, $\tilde{r}_x(f)$ need not be even; (5) Input $\rightarrow$ filter $\rightarrow$ output: $\tilde{r}_y(f) = \tilde{r}_x(f)|H(f)|^2$.

3.5 Important Notation Summary

Symbol	**Meaning**
$x_c(t)$	Real passband signal
$x(t) = x_I(t) + jx_Q(t)$	Complex baseband signal (envelope)
$\tilde{x}(f)$	Fourier transform of $x(t)$
f_c	Carrier frequency
$r_x(\tau)$	Autocorrelation: $\mathbb{E}[x(t+\tau)x^*(t)]$
$\tilde{r}_x(f)$	PSD (Fourier transform of $r_x(\tau)$)
$\tilde{r}_{n_I,n_Q}(f)$	Cross-PSD of I and Q components
$N_0/2$	One-sided WGN power spectral density
τ_i, r_i	Delay and path length of i-th path
$f_{d,i}$	Doppler frequency of i-th path
$\Delta\tau$	Delay spread $= \max_i \tau_i - \min_i \tau_i$
Δf_d	Doppler spread $= \max_i f_{d,i} - \min_i f_{d,i}$
$B_c \approx 1/\Delta\tau$	Coherence bandwidth
$T_c \approx 1/\Delta f_d$	Coherence time
$\mathcal{CN}(0, \sigma^2)$	Circularly symmetric complex Gaussian
$\mathcal{N}(\mu, \sigma^2)$	Real Gaussian distribution
$\mathfrak{Re}\{\cdot\}$, $\mathfrak{Im}\{\cdot\}$	Real and imaginary parts

Good luck on the midterm! Review the worked solutions carefully and practice computing PSD integrals and path geometry.